

12

MEASUREMENT AND SPACE

SIMILAR FIGURES AND SURFACE AREA

OPTIONAL YEAR 9 CORE TOPIC

Figures which have the same shape but are different in size are said to be similar. For example, different-sized photographs of a person where one is an enlargement of the other show the same person, but the size is different. Similar figures are quite useful in calculating the size of objects which are difficult to measure directly, such as the height of a building, the height of a tree or the width of a river.



Chapter outline

		Working mathematically				
12.01	Similar figures*	U	F		R	C
12.02	Properties of similar figures*	U	F		R	C
12.03	Finding sides in similar figures*	U	F	PS	R	
12.04	Surface area of a prism*	U	F	PS	R	
12.05	Surface area of a cylinder*	U	F		R	C

*YEAR 9 EXTENSION

Wordbank

enlargement An increase in the size of a shape

image A transformed shape after it has been enlarged or reduced

matching angles corresponding angles of similar figures in the same position

matching sides corresponding sides of similar figures in the same position

scale factor The amount by which a shape has been enlarged or reduced, equal to $\frac{\text{image length}}{\text{original length}}$

similar To have the same shape but not necessarily the same size, an enlargement or reduction (symbol: ~)

surface area The total area of all the faces of a solid shape

Quiz
Wordbank 12

U = Understanding | **F** = Fluency | **PS** = Problem solving | **R** = Reasoning | **C** = Communicating

Videos (4):

12.03 Finding an unknown side in similar figures

12.04 Surface area of a prism • Surface area of prisms

12.05 Surface area of a cylinder

***Twig* video (1):**

12.04 Cubist art

Quizzes (5):

- Wordbank 12
- SkillCheck 12
- Mental skills 12
- Language of maths 12
- Test yourself 12

Skillsheets (2):

12.03 Finding sides in similar triangles

12.04 Solid shapes

Worksheets (12):

12.01 A page of similar figures • A page of congruent and similar figures

12.02 Enlargements and reductions • Enlarging a logo

12.03 Finding sides in similar figures

12.04 Surface area 1 • Nets of solids • Solid shapes

12.05 Surface area 2 • A page of prisms and cylinders • Applications of area 3

Mind map Similar figures and surface area

Puzzles (4):

12.02 Cartoon enlargement

12.03 Similar triangles • Nested similar triangles

12.05 Car song

Technology (1):

12.04, 12.05 Surface area calculator

Spreadsheets (1):

12.04, 12.05 Surface area calculator

Presentation (1):

12.04 Surface area

Nelson MindTap

To access resources above, visit **cengage.com.au/nelsonmindtap**

In this chapter you will:

(EXTENSION, YEAR 9 CORE)

- identify similar figures and investigate their properties
- use scale factors to find unknown sides in similar figures
- calculate the surface areas of rectangular and triangular prisms
- calculate the surface area of a cylinder

SkillCheck ANSWERS ON P. 617

1 Simplify each fraction.

a $\frac{4}{8}$ **b** $\frac{6}{4}$ **c** $\frac{15}{3}$ **d** $\frac{26}{14}$

2 Evaluate each expression.

a $\frac{2}{3} \times 15$ **b** $\frac{1}{4} \times 32$ **c** $\frac{3}{2} \times 10$ **d** $\frac{5}{7} \times 84$

3 Use a protractor to measure the size of each angle.

a



b



c



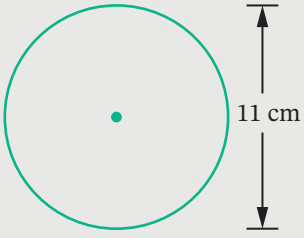
4 Solve each equation.

a $\frac{x}{2} = \frac{10}{4}$ **b** $\frac{y}{40} = \frac{3}{5}$ **c** $\frac{p}{32} = \frac{4}{16}$ **d** $\frac{m}{2} = \frac{12}{18}$

5 Find correct to 2 decimal places this circle's:

a circumference

b area

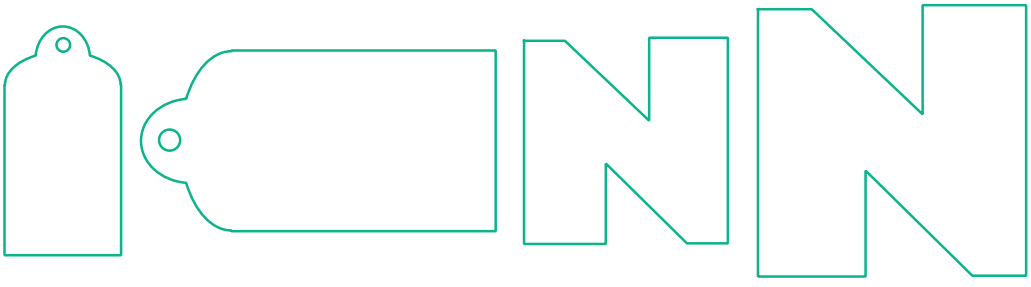


Quiz
SkillCheck 12

12.01

Similar figures 12.01

Similar figures have the same shape but are not necessarily the same size. One figure is an enlargement or reduction of the other figure.



YEAR 9 CORE

Worksheets
A page of similar figures

A page of congruent and similar figures

YEAR 9 CORE

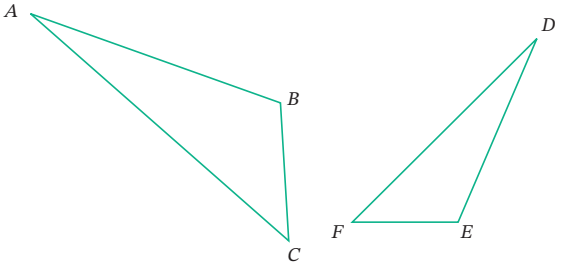
A photograph and its enlargement are examples of similar figures.





iStock.com/USO

Triangles *ABC* and *DEF* below are similar.



We can rotate $\triangle DEF$ anticlockwise to match up the angles so that the triangles look similar.

$\triangle DEF$ is a reduction of $\triangle ABC$.

The symbol for 'is similar to is '~'.

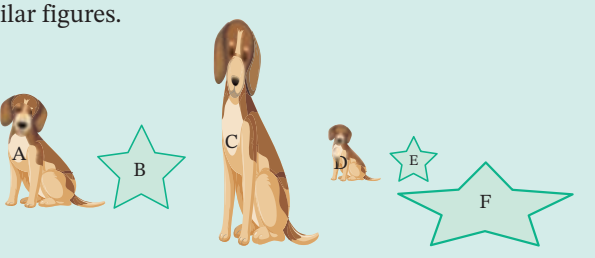
We can write '$\triangle ABC \sim \triangle DEF$', which is read 'Triangle *ABC* is similar to triangle *DEF*'.

When using this notation, we must write the vertices (angles) of the similar figures in matching order:

- $\angle A$ matches $\angle D$
- $\angle B$ matches $\angle E$
- $\angle C$ matches $\angle F$

Example 1

Select 2 pairs of similar figures.



Solution

Figures A and D are similar.

Figures B and E are similar.

They are the same shape but different sizes.

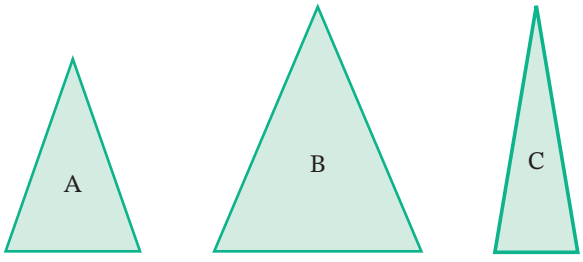
Similar figures U F R C

EXAMPLE 1

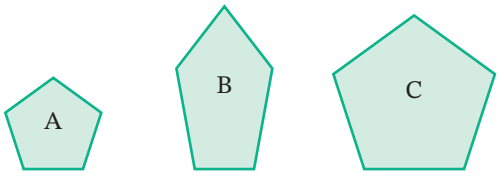
12.01

1 Which pair of shapes are similar? R C

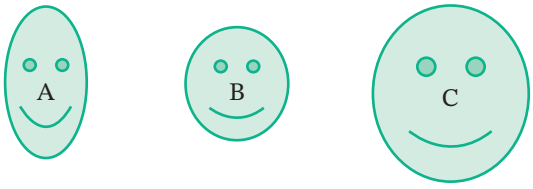
a



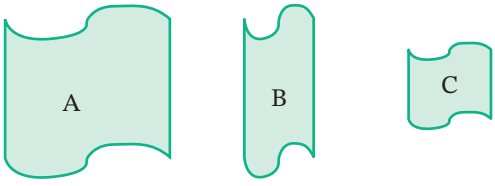
b



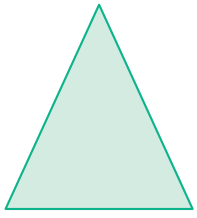
c



d



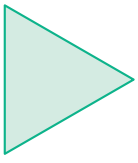
2 Which triangle below is similar to this triangle?
Select **A**, **B**, **C** or **D**.



A



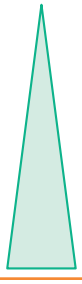
B



C



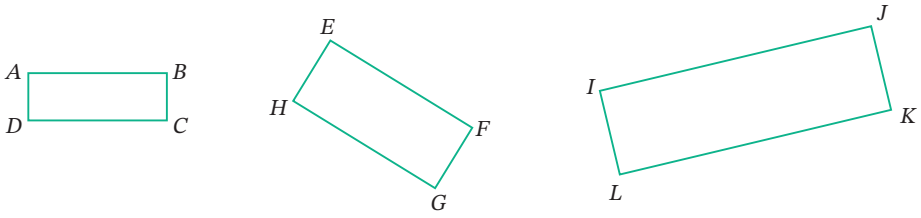
D



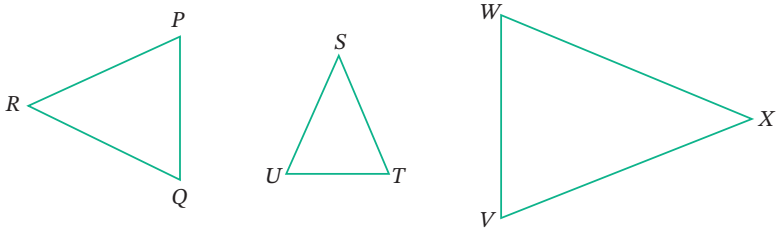
Foundation Standard Complex

3 Find the pair of similar figures in each set of shapes below. Then write a similarity statement of the form '____ ~ ____' naming the vertices in the correct order. R C

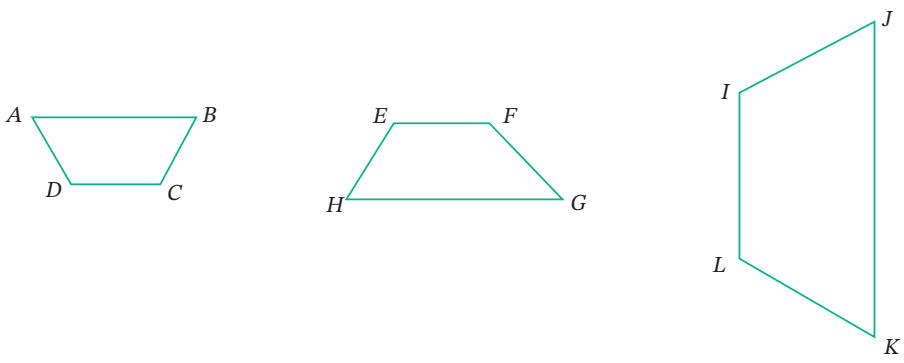
a



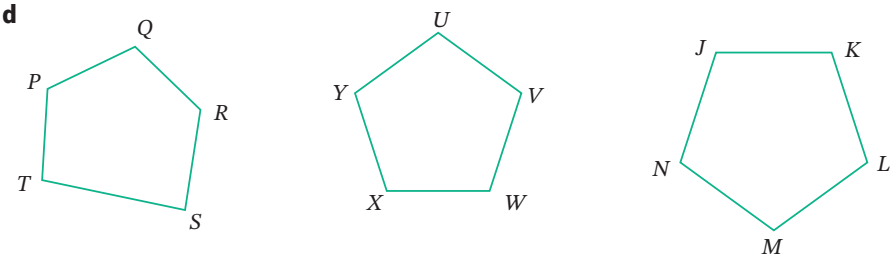
b



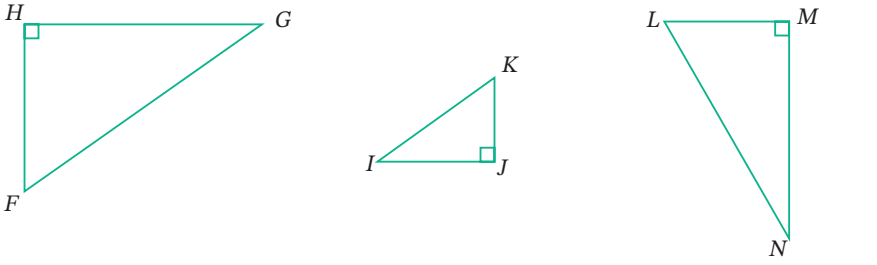
c



d



e

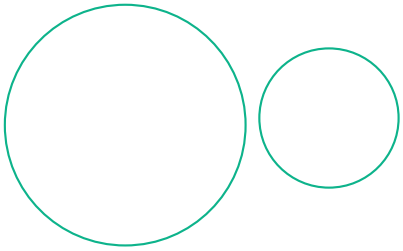


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4 **a** State whether the figures in each pair below are similar. R C

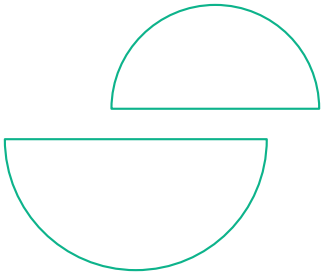
i



ii



iii



b How did you decide on your answer for part **a**?

5 **a** Draw 2 squares that are similar.

b Draw 2 circles that are similar.

c Draw 2 rectangles that are similar.

d Draw 2 similar right-angled triangles.

e Make up your own shape and draw it. Draw another that is similar to it.

6 Discuss these statements and decide whether they are true or false. R C

a All squares are similar.

b All rectangles are similar.

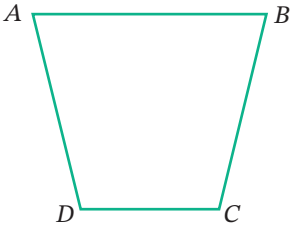
c All triangles are similar.

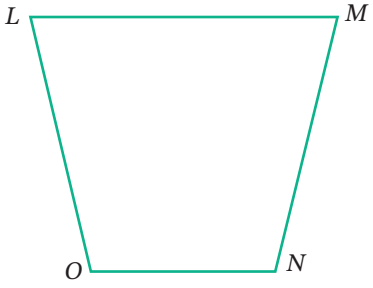
d All circles are similar.

Investigation

Properties of similar figures

You need a ruler and a protractor to complete this investigation. The similar trapeziums below have the same shape but are different in size.





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1 **a** Use a ruler to measure the sides of *ABCD* and the matching sides in *LMNO* in millimetres. Copy and complete this table.

ABCD	*LMNO*	Ratio
$AB =$	$LM =$	$\frac{LM}{AB} =$
$BC =$	$MN =$	$\frac{MN}{BC} =$
$CD =$	$NO =$	$\frac{NO}{CD} =$
$DA =$	$OL =$	$\frac{OL}{DA} =$

b How much larger are all the sides in *LMNO* than the matching sides in *ABCD*?

This number is called the **scale factor**. *ABCD* has been enlarged by the scale factor to get *LMNO*.

2 **a** Use a protractor to measure the angles in each trapezium. Copy and complete this table.

ABCD	*LMNO*
$\angle A =$	$\angle L =$
$\angle B =$	$\angle M =$
$\angle C =$	$\angle N =$
$\angle D =$	$\angle O =$

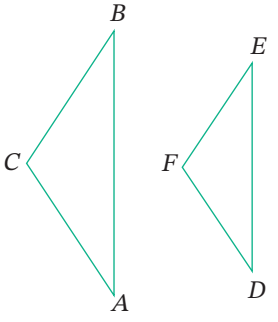
b What do you notice about the sizes of the matching angles?

c Copy and complete the following:

In similar figures

- Matching sides are in the same .....
- Matching angles are .....

3 Look at these similar triangles.



a Measure the angles of each triangle as accurately as you can. Copy and complete this table.

$\angle A =$	$\angle E =$
$\angle B =$	$\angle F =$
$\angle C =$	$\angle G =$

b Measure the sides of each triangle as accurately as you can.
Copy and complete this table.

Length of sides (mm)		Ratio matching sides
$AB =$	$DE =$	$\frac{AB}{DE} =$
$BC =$	$EF =$	$\frac{BC}{EF} =$
$CA =$	$FD =$	$\frac{CA}{FD} =$

Properties of similar figures 12.02

Similar figures

Similar figures have the same shape. One figure is an enlargement of the other figure.

For similar figures:

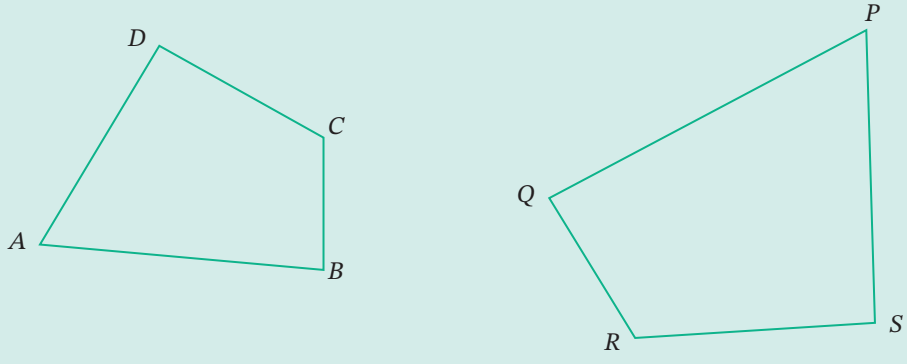
- matching angles are equal
- the lengths of matching sides are in the same ratio

Worksheets
Enlargements and reductions
Enlarging a logo

Puzzle
Cartoon enlargement

Example 2

a Use a protractor to measure the angles in these 2 quadrilaterals.



b Are the matching angles the same size?

c Are the quadrilaterals similar?

Solution

a

Angle	A	B	C	D	P	Q	R	S
Size	80°	90°	115°	75°	80°	90°	115°	75°

b Matching angles are the same size.

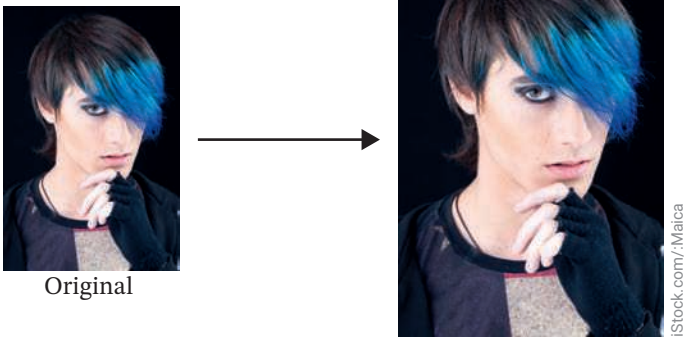
$\angle A = \angle P = 80^\circ$, $\angle B = \angle Q = 90^\circ$, $\angle C = \angle R = 115^\circ$, $\angle D = \angle S = 75^\circ$.

c The quadrilaterals are similar.

The scale factor

A similar figure is an *enlargement* or a *reduction* of another figure. The figure we start with is called the **original**. The enlarged or reduced figure is called the **image**. The scale factor describes the change in size between the original and the image.

For example, this photograph has been enlarged to a bigger size to be placed in a frame, using a scale factor of 1.5. The image is 1.5 times the size of the original.



Original

Image

iStock.com/iMaica

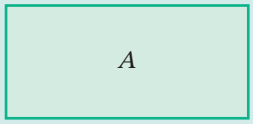
Scale factor

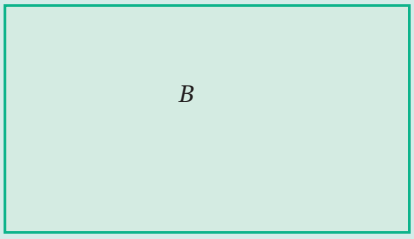
$$\text{Scale factor} = \frac{\text{image length}}{\text{object length}}$$

- A scale factor greater than 1 means that the original has been enlarged
- A scale factor between 0 and 1 means that the original has been reduced

Example 3

These 2 rectangles are similar. By measuring the matching sides, find the scale factor.





In all questions, assume that the left figure is the original and the right figure is the image.

Solution

Length of $A = 3$ cm

Width of $A = 2$ cm

$$\text{Scale factor} = \frac{\text{image length}}{\text{object length}}$$

$$= \frac{6}{3}$$

$$= 2$$

Scale factor is 2.

Length of $B = 6$ cm

Width of $B = 4$ cm

$$\text{Scale factor} = \frac{\text{image length}}{\text{object length}}$$

$$= \frac{4}{2}$$

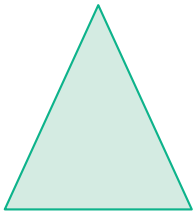
$$= 2$$

EXERCISE 12.02 ANSWERS ON P. 617

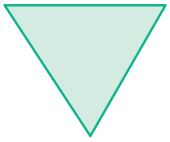
Properties of similar figures U F R C

EXAMPLE 2

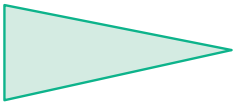
1 By measuring the angles in each figure, determine which figure is similar to this triangle? Select the correct answer **A**, **B**, **C** or **D**.



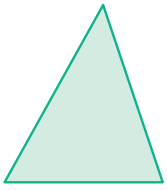
A



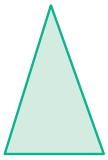
B



C

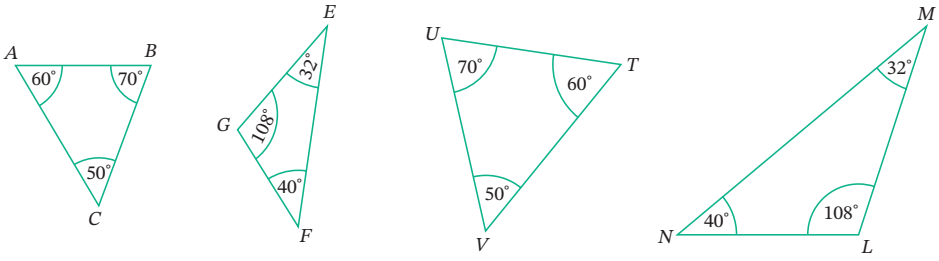


D



Foundation | Standard | Complex

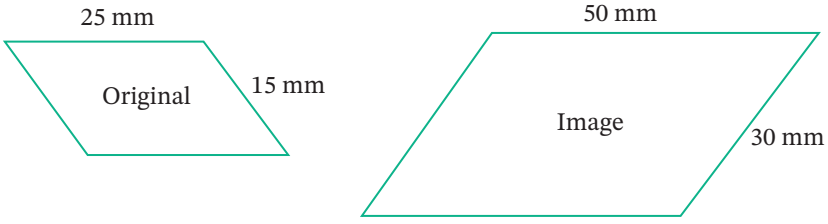
2 Name 2 pairs of similar triangles, writing a similarity statement of the form '____ ~ ____'. Make sure you write the vertices in matching order. R C



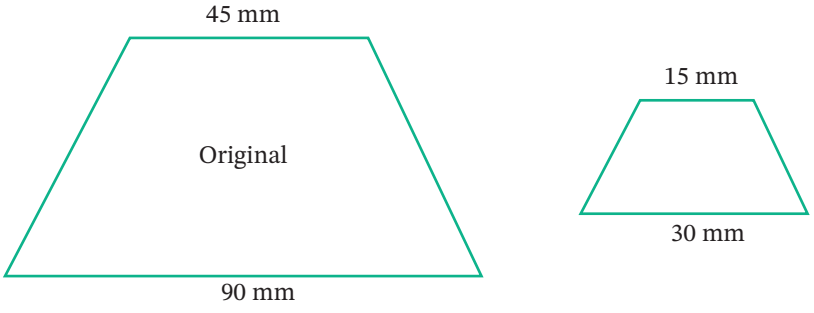
EXAMPLE 3

3 Calculate the scale factor for each pair of similar quadrilaterals. R C

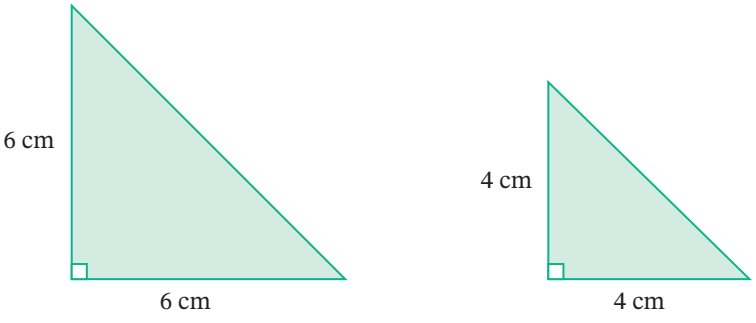
a



b



4 These 2 triangles are similar. R



a Measure all the angles in both triangles. Are matching angles equal?

b Find the ratio of matching sides.

c What is the scale factor?

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5 A photograph is 12 cm long and it is enlarged using a scale factor of 1.5. What is the length of the enlarged photograph? R

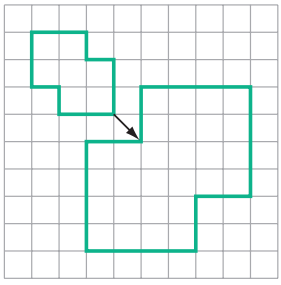
6 Copy each shape and use the given scale factor to enlarge or reduce the figure.

a scale factor of 2 **b** scale factor of $\frac{1}{2}$ **c** scale factor of 1.5

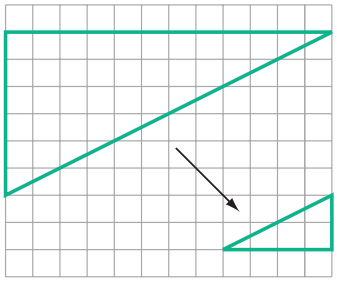


7 By measuring with a ruler, find the scale factor for each pair of similar figures.

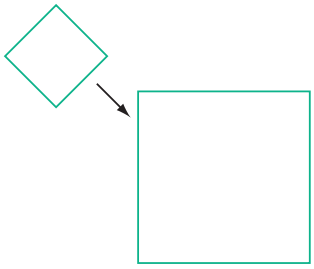
a



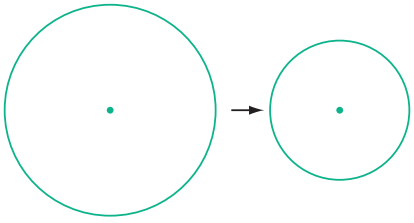
b



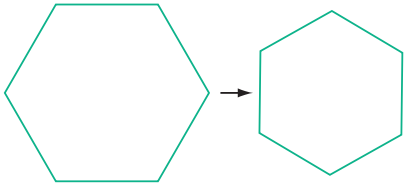
c



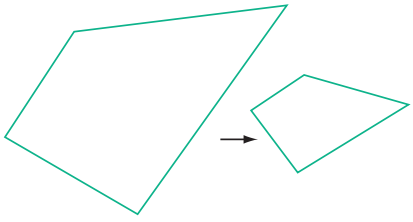
d



e



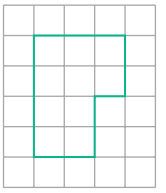
f



Foundation Standard Complex

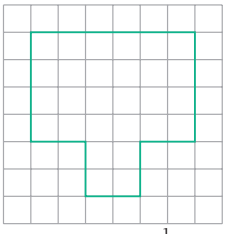
8 Copy each figure and draw its image using the given scale factor.

a



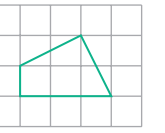
Scale factor 2

b



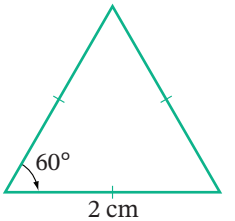
Scale factor $\frac{1}{2}$

c



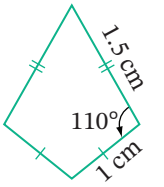
Scale factor 3

d



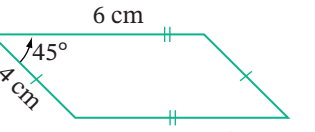
Scale factor 2

e



Scale factor 4

f



Scale factor $\frac{1}{4}$

12.03 Finding sides in similar figures

Video
Finding an unknown side in similar figures

Skillsheet
Finding sides in similar triangles'

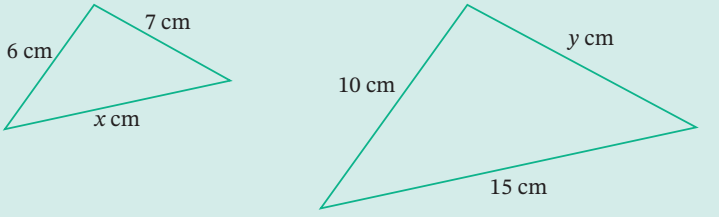
Worksheet
Finding sides in similar figures

Puzzles
Similar figures

Nested similar triangles

Example 4

Find the value of x and y in this pair of similar triangles.



Solution

The sides marked 6 cm and 10 cm are matching sides, and those marked x cm and 15 cm are matching sides.
Since matching sides are in the same ratio:

$$\frac{x}{15} = \frac{6}{10}$$

$$x = \frac{6}{10} \times 15$$
$$= 9$$

Alternative method

Scale factor $= \frac{10}{6} = \frac{5}{3}$

So $x = 15 \div \frac{5}{3} = 9$

y cm and 7 cm are matching sides.

Since matching sides are in the same ratio:

$$\frac{y}{7} = \frac{10}{6}$$

$$y = \frac{10}{6} \times 7$$
$$= 11\frac{2}{3}$$

So $y = 7 \times \frac{5}{3} = 11\frac{2}{3}$

Foundation | Standard | Complex

9780170479516

EXERCISE 12.03 ANSWERS ON P. 617

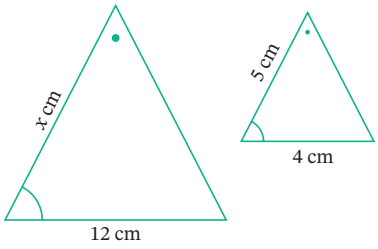
Finding sides in similar figures U F PS R

EXAMPLE 4

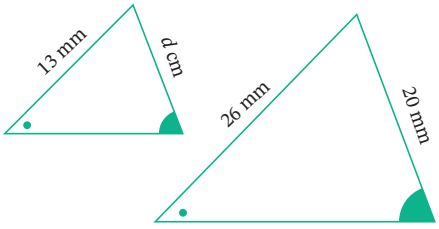
12.03

1 Find the value of the variable in each pair of similar triangles.

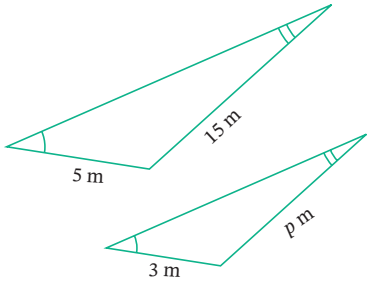
a



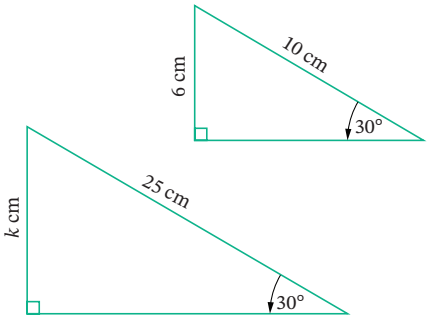
b



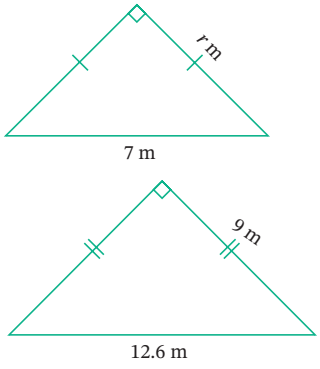
c



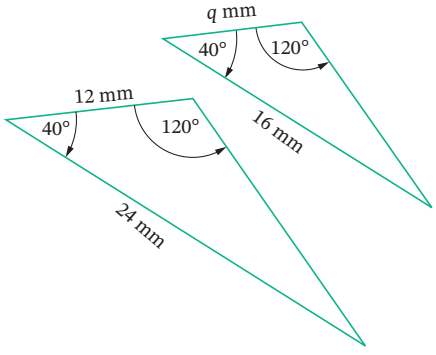
d



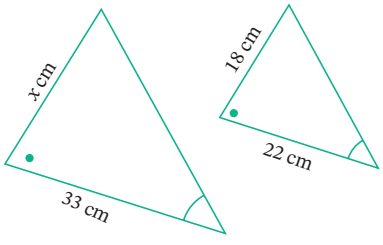
e



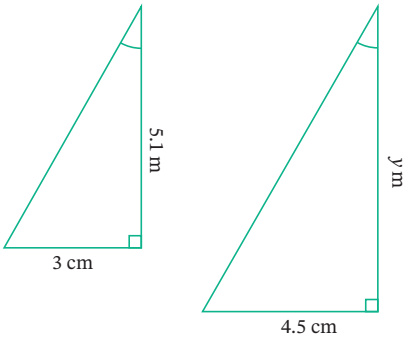
f



g

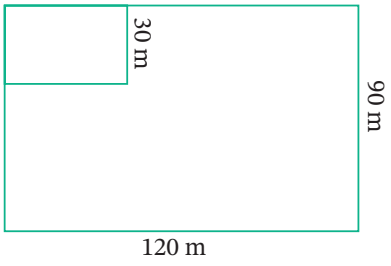


h



Foundation | Standard | Complex

2 A large rectangular paddock, of length 120 m and width 90 m, has a small rectangular pen of width 30 m, in one corner. The paddock and the pen are similar figures.



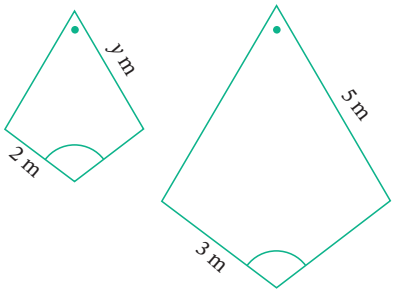
Which of the following is the length of the pen? Select the correct answer **A**, **B**, **C** or **D**.

PS R

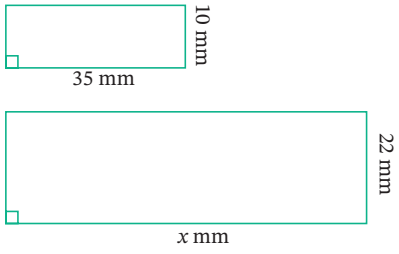
A 22.5 m **B** 37.5 m **C** 40 m **D** 60 m

3 Find the value of the variable in each pair of similar figures.

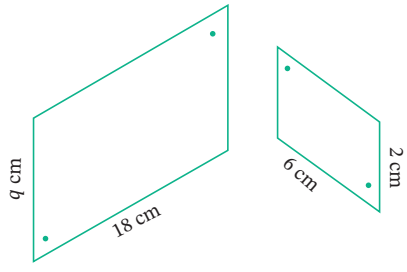
a



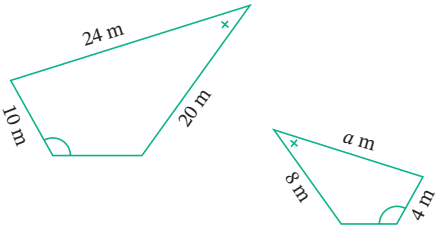
b



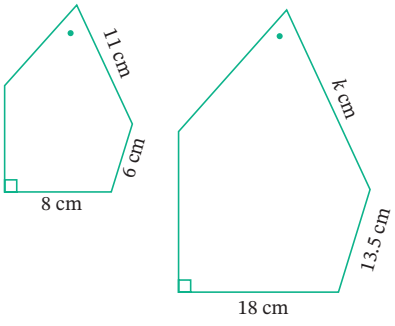
c



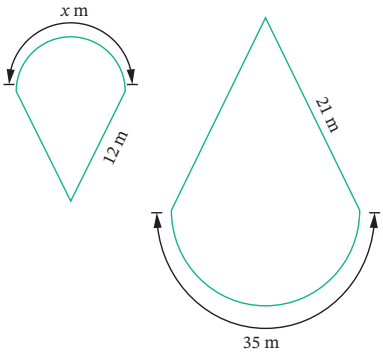
d



e



f

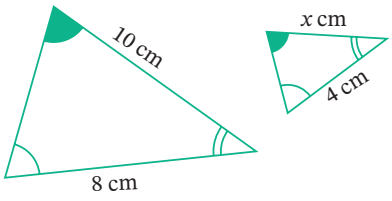


Foundation Standard Complex

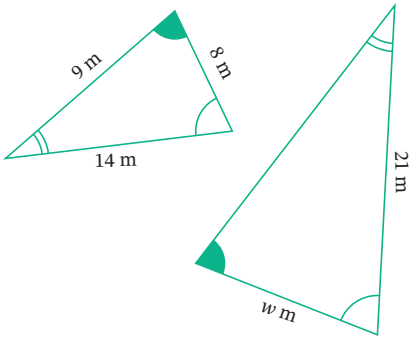
9780170479516

4 For each pair of similar triangles, find the value of the variable.

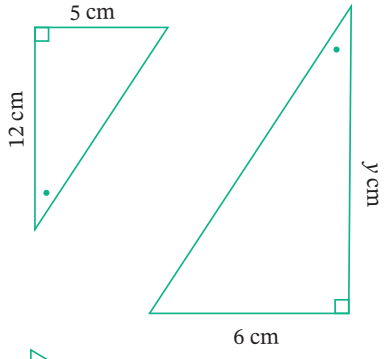
a



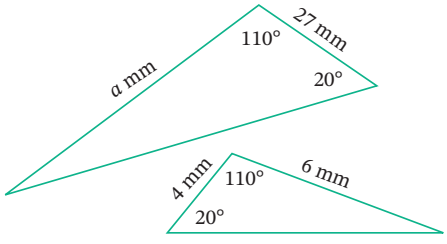
b



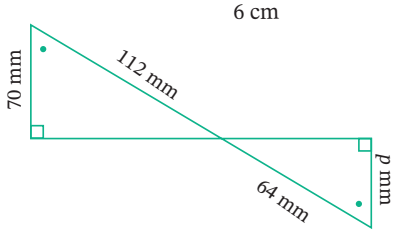
c



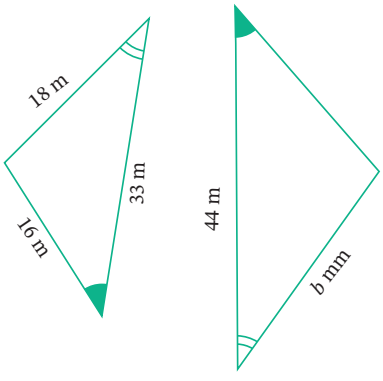
d



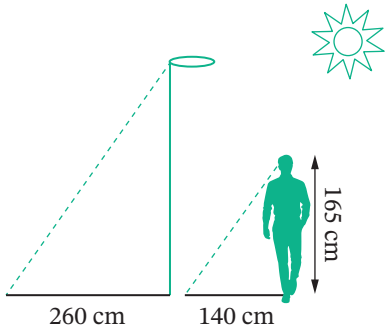
e



f



5 Marcus stands next to a pole supporting a netball ring on a sunny day. He is 165 cm tall and casts a shadow of 140 cm. The pole casts a shadow of 260 cm. How high is the pole, correct to the nearest centimetre? PS R



Foundation Standard Complex

Quiz
Mental skills 12

Mental skills 12: Maths without calculators ANSWERS ON P. 617

Time differences

1 Study each example.

a What is the time difference between 11:40 a.m. and 6:15 p.m.?

From 11:40 a.m. to 5:40 p.m. = 6 hours

Count: '11:40, 12:40, 1:40, 2:40, 3:40, 4:40, 5:40'

From 5:40 a.m. to 6:00 p.m. = 20 min

From 6:00 p.m. to 6:15 p.m. = 15 min

5 hours + 20 min + 15 min = 6 hours 35 min

OR:



b What is the time difference between 1645 and 2320?

From 1645 to 2245 = 6 hours (22 − 16 = 6)

From 2245 to 2300 = 15 min

From 2300 to 2320 = 20 min

6 hours + 15 minutes + 20 minutes = 6 hours 35 minutes

OR:



2 Now find the time difference between:

a 11:10 a.m. and 7:40 p.m.

b 6:20 p.m. and 12:00 midnight

c 4:45 p.m. and 8:10 p.m.

d 2:35 a.m. and 10:50 a.m.

e 1:05 p.m. and 12:30 a.m.

f 9:35 a.m. and 11:15 a.m.

g 0425 and 0935

h 1440 and 2025

i 7:55 a.m. and 3:50 p.m.

j 2:40 p.m. and 10:20 p.m.

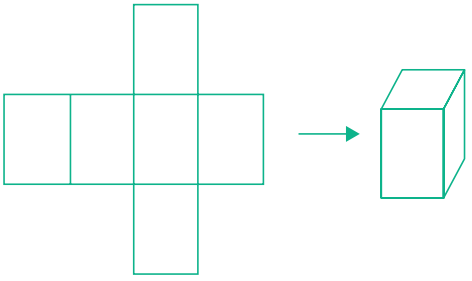
9780170479516

Surface area of a prism

12.04

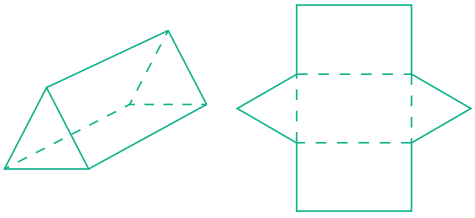
The **surface area** of a solid is the total area of all the faces of the solid.

The surface area of a solid can be easily calculated by first drawing the net of the solid, finding the area of each face, then adding the areas together.



A rectangular prism and its net.

The prism has 6 faces.



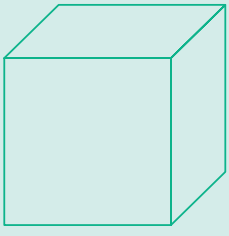
A triangular prism and its net.

The prism has 5 faces.

Example 5

Find the surface area of each prism.

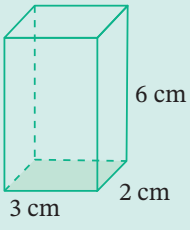
a



Cube

side length 5 cm

b



Rectangular prism

Solution

a A cube has 6 identical square faces.

$$\text{Surface area} = 6 \times 5^2 \qquad 6 \times \text{area of a square}$$
$$= 150 \text{ cm}^2$$

Surface area is an area, so it is measured in square units such as cm^2 or m^2.

Videos
Surface area of a prism
Surface area of prisms
Cubist art

Worksheets
Surface area 1
Nets of solids
Solid shapes

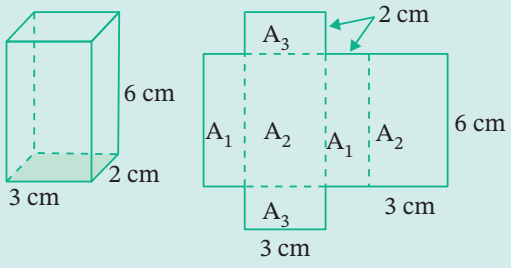
Skillsheet
Solid shapes

Presentation
Surface area

Technology
Surface area calculator

Spreadsheet
Surface area calculator

b



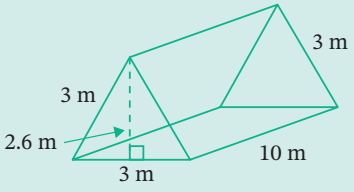
A rectangular prism has 6 rectangular faces. Opposite faces are the same size, such as the top and bottom, shown as A_3 in the diagram.

$$\text{Surface area} = 2 \times A_1 + 2 \times A_2 + 2 \times A_3 \qquad \text{2 side faces + 2 front/back faces}$$
$$= 2 \times (6 \times 2) + 2 \times (6 \times 3) + 2 \times (2 \times 3) \qquad \text{+ 2 top/bottom faces}$$
$$= 72 \text{ cm}^2$$

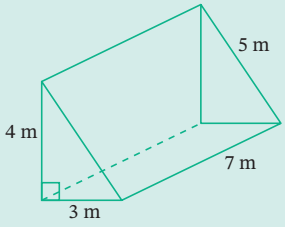
Example 6

Find the surface area of each prism.

a

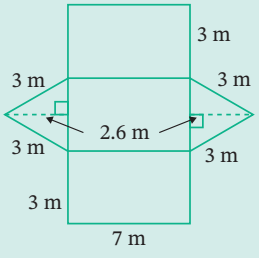


b



Solution

a This triangular prism has 5 faces. Its base is an equilateral triangle with side length 3. Its faces are made of 2 identical triangles (front and back) and 3 equal rectangles (sides and bottom).



$$\text{Surface area} = 2 \times \left(\frac{1}{2} \times 3 \times 2.6\right) + 3 \times (10 \times 3) \qquad \text{2 front/back faces + 3 rectangular faces}$$
$$= 97.8 \text{ m}^2$$

b This triangular prism has 5 faces: 2 identical triangles and 3 different rectangles of length 7 m.

$$\text{Surface area} = 2 \times \left(\frac{1}{2} \times 3 \times 4\right) + (3 \times 7) + (4 \times 7) + (5 \times 7) \qquad \text{2 front/back faces + bottom + left + right}$$
$$= 96 \text{ m}^2$$

EXERCISE 12.04 ANSWERS ON P. 617

Surface area of a prism U F PS R

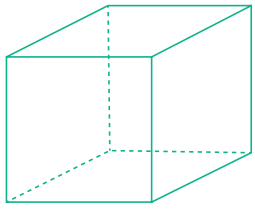
EXAMPLE 5

12.04

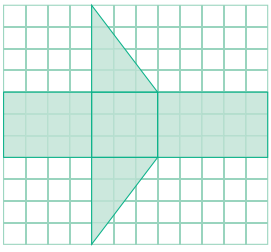
1 Find the surface area of a cube with side length 6 cm.
Select the correct answer **A**, **B**, **C** or **D**.

A 24 cm^2 **B** 144 cm^2

C 36 cm^2 **D** 216 cm^2

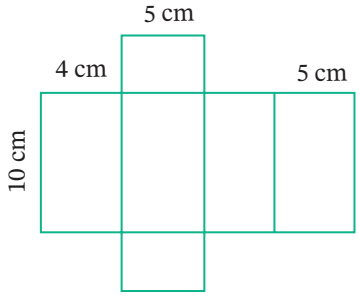


2 Name and draw the solid that this net would make.

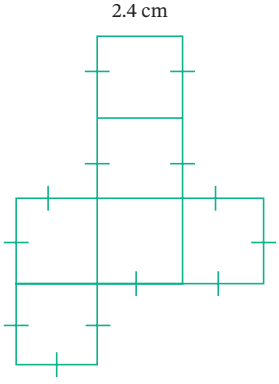


3 Name the solid that can be made from each net and find its surface area.

a

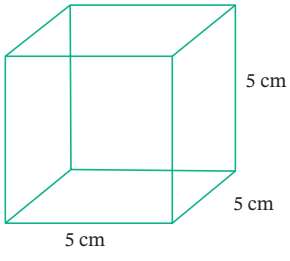


b

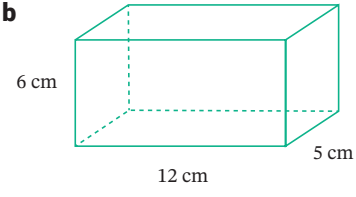


4 Find the surface area of each prism.

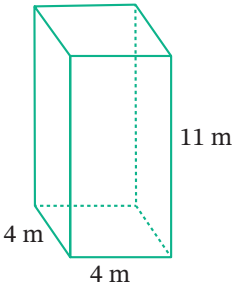
a



b



c

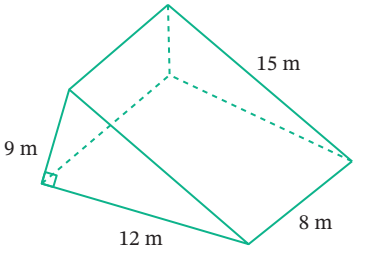


Foundation Standard Complex

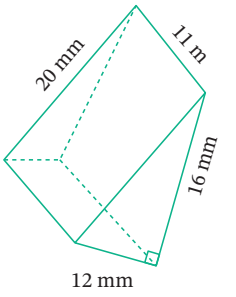
EXAMPLE 6

5 Find the surface area of each triangular prism.

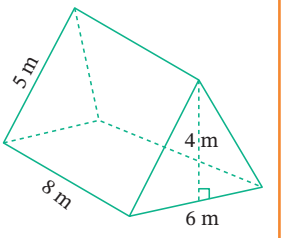
a



b

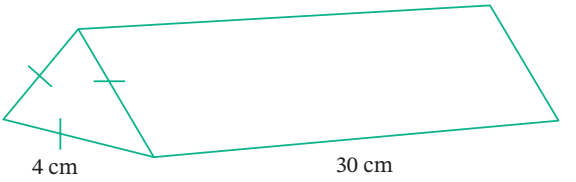


c



6 Bodhi bought a ruler in the shape of a triangular prism as shown in the diagram. If the ruler is to be covered with aluminum foil, calculate the surface area to be covered.

PS R



height of triangle = 3.5 cm

7 Mary purchased a metal container and wishes to paint all the sides except the floor. PS R

a Calculate the surface area to be painted.

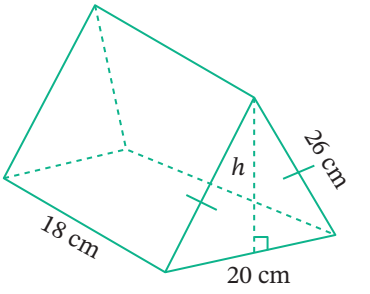
b If a 4-litre can of paint can cover 40 square metres, how many cans does Mary need to complete her task?



Shutterstock.com/Cathetus

8 **a** Find the perpendicular height, h, of this triangular prism, correct to 2 decimal places.

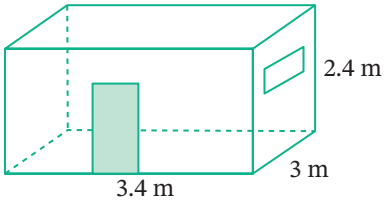
b Calculate the total surface area of the prism.



Foundation Standard Complex

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9 Duc wishes to tile his new bathroom to a height of 1.2 m. The door is 0.9 m wide and the window is higher than 1.2 m. PS R

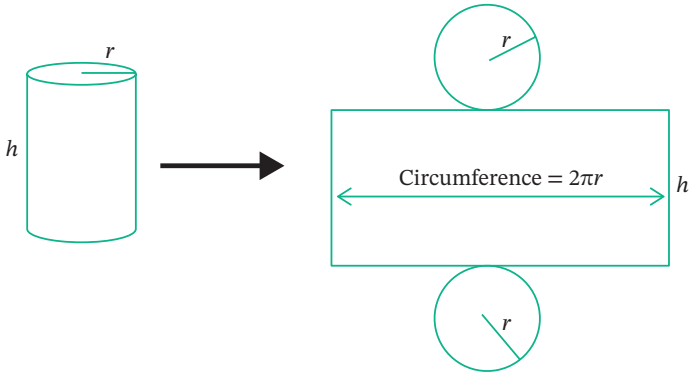


a Calculate how many square metres of tiles Duc needs? Answer to the nearest whole metre.

b If the tiles cost $65 per square metre and the tiler charges $60 per square metre, find the total cost for tiling the bathroom.

Surface area of a cylinder 12.05

When a closed cylinder is opened out flat, its net is a rectangle and 2 identical circles:



The curved surface of the cylinder is the rectangle whose length is the circumference of the cylinder.

Surface area of a cylinder = area of 2 circles + area of rectangle

$$= 2 \times \pi r^2 + 2\pi r \times h$$

$$= 2\pi r^2 + 2\pi rh$$

Surface area of a closed cylinder

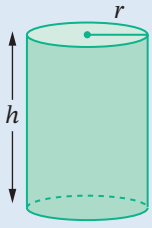
$SA = 2\pi r^2 + 2\pi rh$

where r = radius of circular base

h = perpendicular height

The area of the 2 circular ends = $2\pi r^2$

and the area of the curved surface = $2\pi rh$.



Video
Surface area of a cylinder

Worksheets
Surface area 2

A page of prisms and cylinders

Applications of area 3

Puzzle
Car song

Technology
Surface area calculator

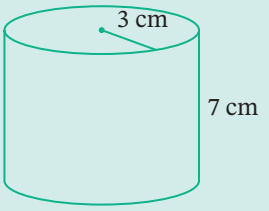
Spreadsheet
Surface area calculator

Foundation Standard Complex

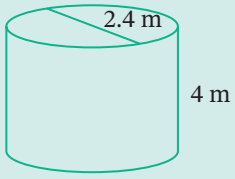
Example 7

Find correct to one decimal the surface area of each cylinder.

a



b This water tank is open at the top.



Solution

a $SA = 2 \times \pi \times 3^2 + 2 \times \pi \times 3 \times 7$

$= 188.4955 \cdots$

$\approx 188.5 \text{ cm}^2$

b Diameter $= 2.4$ m, so $r = \frac{1}{2} \times 2.4 = 1.2$ m

$SA =$ area of 1 circle + area of curved surface

$= \pi \times 1.2^2 + 2 \times \pi \times 1.2 \times 4$

$= 34.6831 \cdots$

$= 34.7 \text{ m}^2$

EXERCISE 12.05 ANSWERS ON P. 618

Surface area of a cylinder U F PS R C

EXAMPLE 7

1 Find the surface area of a closed cylinder with radius 15 mm and height 20 mm, correct to 1 decimal place. Select the correct answer **A**, **B**, **C** or **D**.

A 3298.7 mm^2 **B** 1979.2 mm^2 **C** 2591.8 mm^2 **D** 28734.3 mm^2

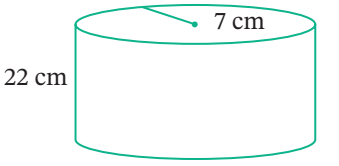
2 For this cylinder, find correct to 2 decimal places:

a the circumference of the circular base

b the area of the circular base

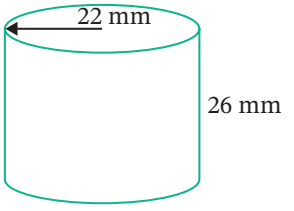
c the area of the curved surface

d the total surface area of the closed cylinder

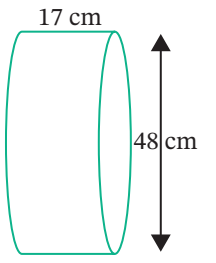


3 Find the surface area of each cylinder, correct to one decimal place.

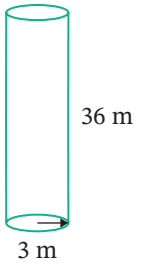
a



b



c

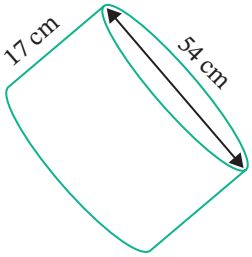


Foundation Standard Complex

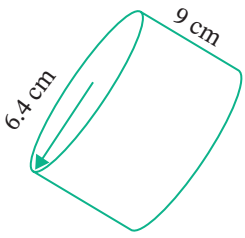
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4 Find the surface area of each cylinder, which are open at the top. Answer to one decimal place. R C

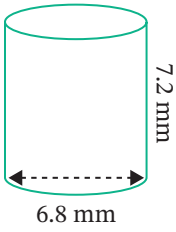
a



b

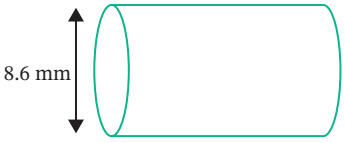


c

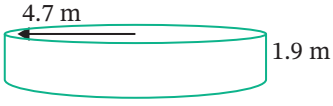


5 Find the area of the curved surface of each cylinder correct to 2 decimal places.

a

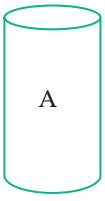


b



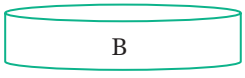
6 Find correct to the nearest square centimetre the external (outside) surface area of an open-ended pipe 3 m long and with a 3 cm radius. R C

7 Which of these cylinders has the greater surface area?



Radius 4 cm

Height 12 cm



Radius 12 cm

Height 4 cm

8 An aeroplane hangar has the shape of half a cylinder with one open end.

The diameter of the hangar is 12 m.

The length of the hangar is 15 m.

Find its external surface area, correct to 2 decimal places.

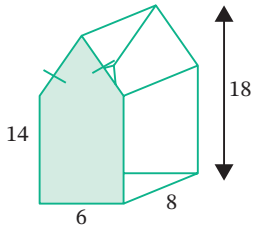


Shutterstock.com/photographer DmitryStrong

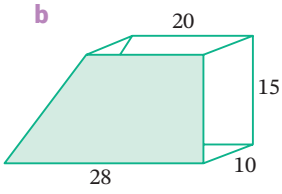
Power plus ANSWERS ON P. 618

1 Find the surface area of each prism. All measurements are in centimetres.

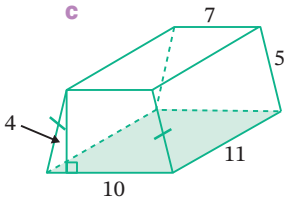
a



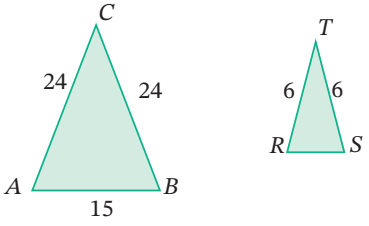
b



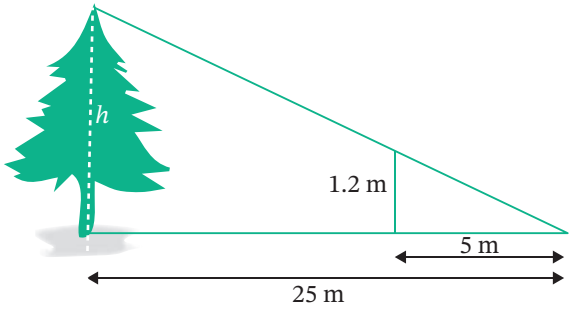
c



2 $\triangle ABC$ and $\triangle RST$ are similar triangles. How many times is the area of $\triangle ABC$ the size of the area of $\triangle RST$?

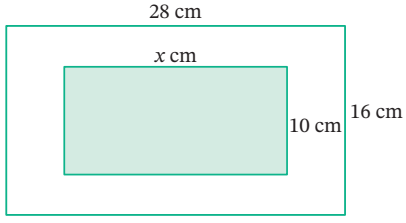


3 A tree casts a shadow of 25 m long on the ground. At the same time a 1.2 m stick casts a shadow of 5 m.

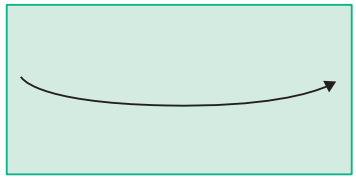


Find the height of the tree, correct to one decimal place.

4 These 2 rectangles are similar. How many times is the area of the large rectangle as large as the area of the small rectangle?



5 Toula has a metal sheet 30 cm long and 20 cm high. She curls it to form a tube. Find the diameter of the tube, correct to one decimal place.



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CHAPTER 12 REVIEW

Quiz
Language of maths 12

Language of maths

circumference	cube	curved surface	cylinder
enlargement	face	image	net
original	perpendicular height	pi (π)	prism
radius	ratio	rectangular prism	reduction
scale factor	similar (~)	surface area	triangular prism

1 Find 2 different meanings for the word 'scale'.

2 How is the geometrical meaning of 'similar' different from its everyday meaning?

3 What is $\frac{\text{image length}}{\text{object length}}$?

4 What can you say about the value of the above fraction if a figure is reduced in size?

5 How many faces has a triangular prism?

6 What shape is the curved surface of a cylinder?

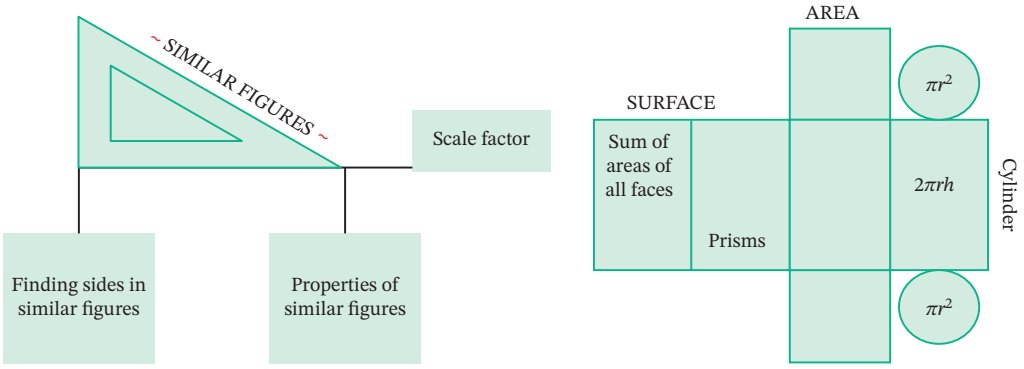
Topic summary

Worksheet
Mind map: Similar figures and surface area

- Write in your own words what you have learnt about similar figures and surface area.
- What parts of this topic did you have difficulty with? Discuss them with a friend.
- Where might you use the skills acquired in this chapter?

Print (or copy) and complete these mind maps of the topics, adding detail to its branches and using pictures, symbols and colour where needed. Ask your teacher to check your work.

SIMILAR FIGURES AND SURFACE AREA



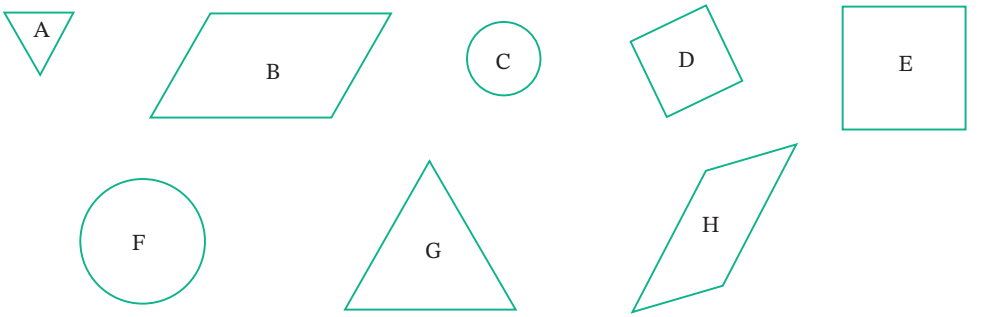
Quiz
Test yourself 12

TEST YOURSELF 12

ANSWERS ON P. 618

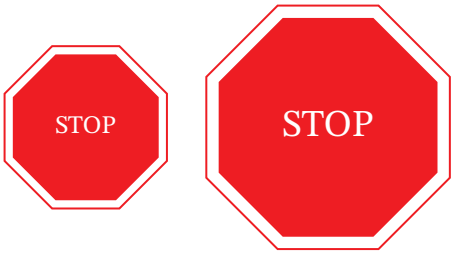
12.01

1 Match pairs of similar figures.



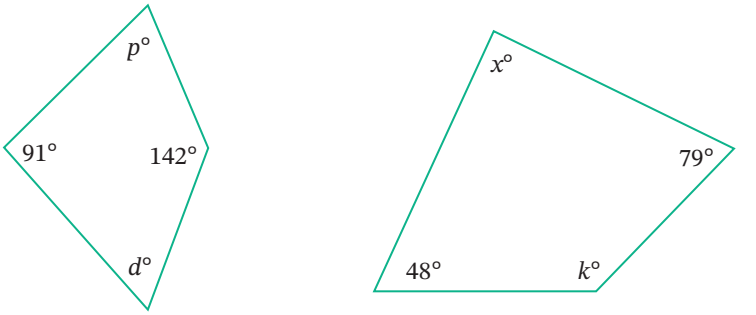
12.02

2 By measuring the matching sides, find the scale factor of these similar figures.



12.02

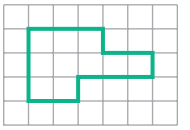
3 Find the value of all variables in these 2 similar quadrilaterals.



12.02

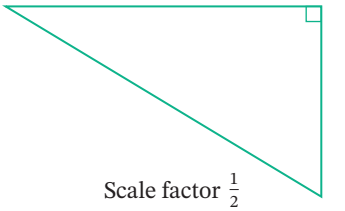
4 Draw the image of each figure using the given scale factor.

a



Scale factor 2

b



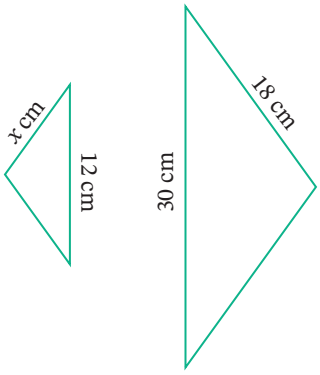
Scale factor $\frac{1}{2}$

Foundation | Standard | Complex

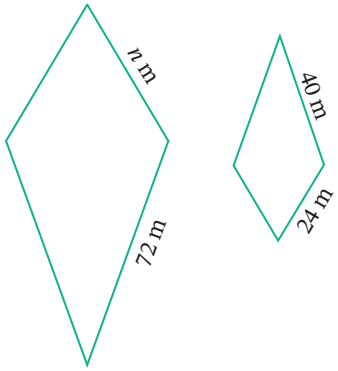
9780170479516

5 Find the scale factor for each pair of similar figures. 12.03

a



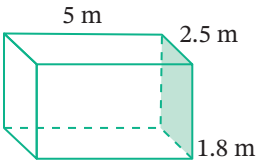
b



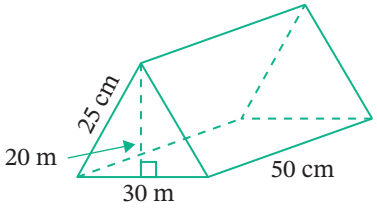
6 Find the value of the variable in each pair of similar figures above. 12.03

7 Find the surface area of each prism. 12.04

a

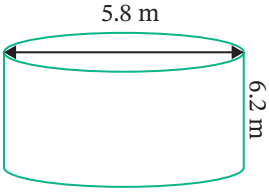


b

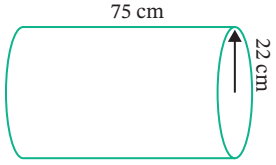


8 Find the surface area of each cylinder, correct to one decimal place. 12.05

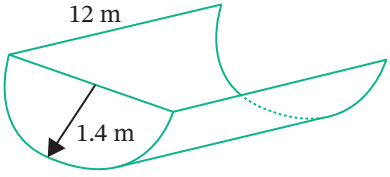
a



b



9 Find the surface area of this half-cylinder that is open at one end. Answer correct to 2 decimal places. 12.05



Foundation | Standard | Complex

PRACTICE SET 4 ANSWERS ON P. 618

10.01

1 Simplify each ratio.

a 2 : 6

b 96 : 34

c 2 m to 15 cm

d 10 minutes to 2 hours

e $\frac{2}{5} : \frac{1}{3}$

f 0.7 : 2.8

10.06

2 For each pair of sizes, find the better buy.

a Chips: 300 g for $5.20 or 400 g for $7.20

b Pasta: 1.5 kg for $3.30 or 750 g for $1.80

11.01

3 Complete this table of values using the formula $y = 2x + 9$.

x	−1	0	1	2	3
y					

10.02

4 The ratio of boys to girls at a conference is 4 : 3. If there are 15 girls, find the number of boys.

10.03

5 **a** If 5 cm on a map represents 1 km in actual distance, what length would be used to represent 5 km?

b Using a scale of 1 : 500, what actual length, in metres, would a scaled length of 4 cm represent?

10.08

6 **a** A train travels 264 km in 8 hours. What is its average speed?

b A bus travels at 45 km/h for 5 hours. How far does it travel?

11.02

7 Find the formula for this table of values.

x	2	3	4	5	6	7
y	6	9	12	15	18	21

11.03

8 Here are the first 3 shapes in a pattern made of toothpicks.



a Copy and complete this table.

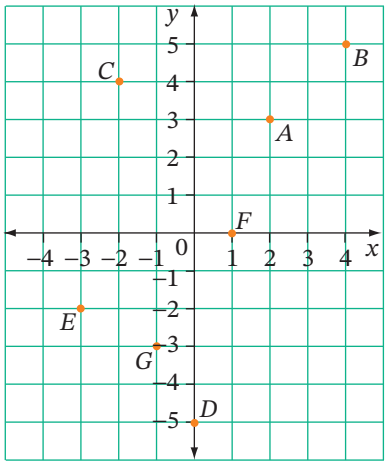
Number of triangles t	1	2	3	4	5	8
Number of toothpicks, p	3					

b Write the rule for the pattern in words.

c Write the rule as a formula.

d How many toothpicks are needed to build 20 triangles?

9 Write the coordinates of each point marked on the number plane. 11.04



10 Graph each linear equation on a number plane after copying and completing its table of values. 11.06

a $y = x + 3$

x	−1	0	1	2
y				

b $y = 4x - 1$

x	−1	0	1	2
y				

11 A truck is loaded with fruit and vegetables in the ratio 5 : 9. The total mass is 4.2 tonnes. What is the mass of the fruit carried in the truck? 10.04

12 In 20 hours, a train travels 800 km. How far would it travel in 11 hours at the same rate? 10.08

13 Test whether the point (3, 7) lies on the line with the equation: 11.06

a $y = 2x + 4$

b $y = 3x - 2$

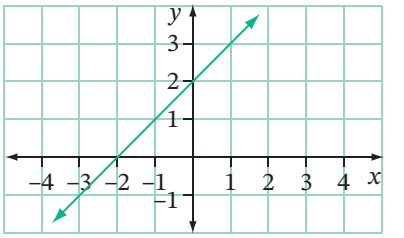
14 Water flows into a tank at a rate of 15 L per hour. How long will it take to fill a tank that holds 750 L? 10.07

15 Graph the line $y = 2$ on a number plane. 11.06

16 Graph the line $y = 2x - 3$ and use it to solve the equation $2x - 3 = 5$ graphically. 11.12

11.07

17 Find the equation of this line.



10.09

18 Convert

a 15 m/s to m/h

b 3 tonnes/day to kg/day

c 30 L/minute to mL/s

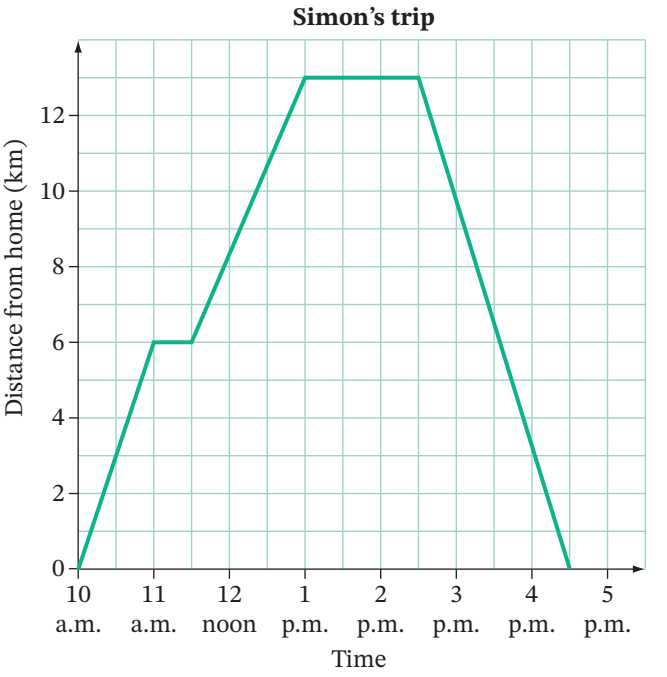
d 60 km/h to m/s

11.13

19 Graph the lines $y = 2x$ and $y = x + 2$ on the same number plane and find their point of intersection.

10.10

20 Simon's cycling trip is described by this travel graph.



a When, and for how long, did Simon stop for lunch?

b How far did he cycle in total?

c What was Simon's speed on the return journey home?

11.12

21 Solve the equation $3x + 2 = x - 1$ graphically.

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22 Emilia planted a rose bush in her garden. The bush was 20 cm tall when it was first planted and grows 3 cm per month. 11.11

a Copy and complete this table, where h is the height of rose bush in centimetres at t months.

Time, t months	0	2	4	6	8	10	12
Height, h cm	20	26					

b Graph this table of values as a line on a number plane.

c Find the equation of the line.

d Find the height of the rose bush after 5 months.

e When will the height reach 74 cm?

23 Leo and Chris share the cost of their internet bill in the ratio of 3 : 5. What fraction does Chris pay? Select the correct answer **A**, **B**, **C** or **D**. 10.02

A $\frac{2}{5}$ **B** $\frac{5}{8}$

C $\frac{5}{2}$ **D** $\frac{1}{5}$

24 Dounya takes 24.5 seconds to walk 70 metres, while Anna walks 160 metres in 1 minute. Answer correct to one decimal place. 10.08

a What is the average speed of each person?

b If each person walks 2 km at their normal speed, who finishes first, and by how much?

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